

Programme	B. Sc. Mathematics Honours			
Course Code	MAT5EJ303 (2)			
Course Title	CONVEX OPTIMIZATION			
Type of Course	Elective (Specialisation- Data Science)			
Semester	V			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Linear Algebra and Multi Variable Calculus			
Course Summary	The course covers the basic theory of convex sets and functions, optimization theory of convex functions and Lagrangian duality. The concepts explored in this course are important for data science, as they underpin many algorithms and methods in machine learning, optimization, and statistical analysis. For instance, understanding gradients and Hessians is essential for optimizing cost functions, while knowledge of convex optimization is vital for developing efficient algorithms. This mathematical foundation will enable data scientists to design, analyse, and implement sophisticated models and solutions.			

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Prove the basic properties of convex sets and functions.	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Model simple problems using convex optimization methods and solve them.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Formulate the dual of a convex optimization problem and describe the properties.	U	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		1. K. G. Binmore, Mathematical Analysis: A straightforward approach, 2nd edition, Cambridge University Press, 1982. 2. Stephen Boyd, and Lieven Vandenberghe. Convex optimization. Cambridge university press, 2004.		
Module	Unit	Content	Hrs (48+12)	Ext. Marks (70)
I	Review of Multivariable Calculus		10	Min 15
	1	Scalar and vector fields - Directional and Partial Derivatives		
	2	Differentiable functions and total Derivative - Matrix representation - Gradient and Jacobian		
	3	Chain rule for differentiation - matrix form		
	4	Stationary points - conditional for stationarity		
	5	Second derivatives and Hessian Matrix.		
	6	Mean value theorems, second order Taylor's theorem		
	7	Eigenvalues of Hessian		
	8	Classification of stationary points.		
	Chapter 19 of Text Book 1 - pages 190-231.			
II	Convexity		14	Min 15
	9	Affine and Convex Sets		
	10	Convexity preserving operations		
	11	Generalized inequalities		
	12	Supporting and separating hyperplanes		
	13	Dual cones and generalized inequality		
	14	Basic properties and examples of convex functions		
	15	Convexity preserving operations		
	16	Quasi convex, log convex functions		
	17	Convexity and generalized inequalities		
	Chapter 2 and 3 of Text Book 2.			
III	Convex Optimization Problems		12	
	18	Optimization problems and convex optimization		

	19	Linear optimization problems		Min 15
	20	Quadratic optimization problems		
	21	Geometric programming		
	22	Generalized inequality constraints		
	19	Vector optimization		
	Chapter 4 of Text Book 2			
IV	Duality		12	Min 15
	20	The Lagrange dual function		
	21	The Lagrangian dual and geometric interpretation		
	22	Saddle point interpretation		
	23	Optimality condition		
	24	Theorems of alternatives		
	25	Generalized inequalities		
	Chapter 5 of Text Book 2			
V (Open Ended)	Open Ended		12	
	27	Instances of practical problems that can be solved with convex optimization methods discussed in the course such as linear classifiers, support vector machines, linear and logistic regression.		
References:				
1. David G. Luenberger and Yinyu Ye. Linear and nonlinear programming. 4th edition. Springer, 2015.				
2. Niels Lauritzen, Undergraduate Convexity: From Fourier And Motzkin To Kuhn And Tucker, World Scientific, 2013.				

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	0	2	3	2	3	2	3	1	2
CO 2	2	3	1	2	3	2	3	3	3	1	3
CO 3	2	2	0	3	3	2	3	2	3	1	2

Correlation Levels:

Level	Correlation
2-3	1N2il
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓